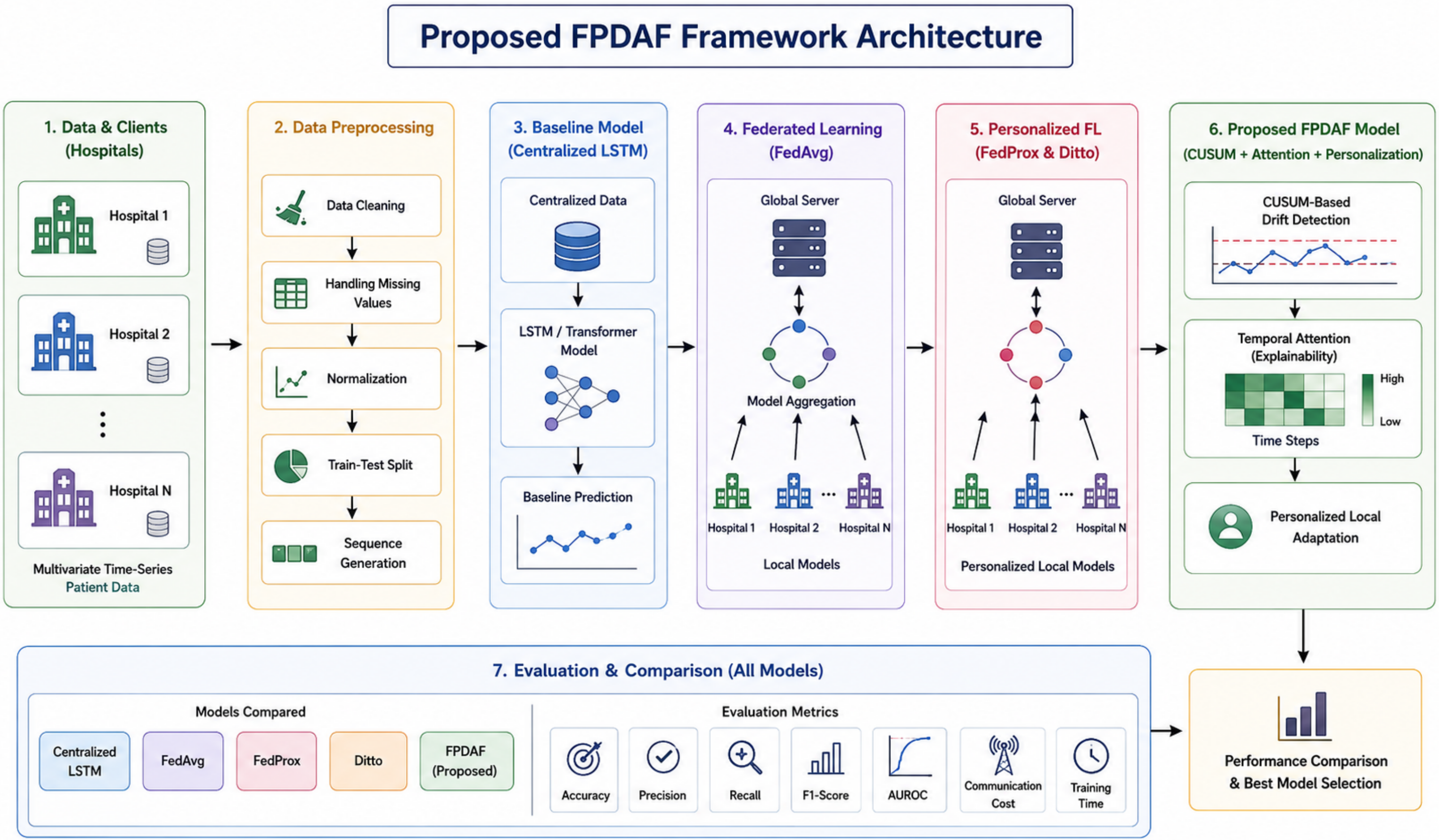
Proposed FPDAF Framework Architecture
1. Data & Clients (Hospitals)
Hospital 1
Hospital 2
Hospital N
Multivariate Time-Series Patient Data
2. Data Preprocessing
Data Cleaning
Handling Missing Values
Normalization
Train-Test Split
Sequence Generation
3. Baseline Model (Centralized LSTM)
Centralized Data
LSTM / Transformer Model
Baseline Prediction
4. Federated Learning (FedAvg)
Global Server
Model Aggregation
Hospital 1
Hospital 2
Hospital N
Local Models
5. Personalized FL (FedProx & Ditto)
Global Server
Hospital 1
Hospital 2
Hospital N
Personalized Local Models
6. Proposed FPDAF Model (CUSUM + Attention + Personalization)
CUSUM-Based Drift Detection
Temporal Attention (Explainability)
High
Low
Time Steps
Personalized Local Adaptation
7. Evaluation & Comparison (All Models)
Models Compared
Centralized LSTM
FedAvg
FedProx
Ditto
FPDAF (Proposed)
Evaluation Metrics
Accuracy
Precision
Recall
F1-Score
AUROC
Communication Cost
Training Time
Performance Comparison & Best Model Selection